

Reply to: “Application of digital pathology in liver transplantation”

To the Editor:

We appreciated the letter from *Li and colleagues*¹ regarding our recently published study in the *Journal of Hepatology*.² The authors raised relevant points that allow us to discuss the use of digital pathology in liver transplantation, especially for donor liver steatosis assessment. With this letter, we want to clarify these points and provide insights into this evolving field.

First, *Li and colleagues* suggested that our 292-liver-biopsy sample size was small, particularly if compared to the 8,836 colorectal cancer tumor samples and 1,225+773 gastric cancer patients used to develop other deep learning tools. We acknowledge the importance of representative sample size, but we also believe that each study should be contextualized in its appropriate clinical scenario. The field of transplantation, in particular liver transplantation, cannot match the numbers of the oncological domain. As a reference, the overall number of liver transplants performed worldwide in 2022 was 37,436.³ In the same year, the new cases of colorectal cancer and gastric cancer were 1,926,118 and 968,350, thus resulting in the third and fifth most common cancers in the world.⁴ A brief literature review focused on deep learning models/artificial intelligence approaches pertinent to donor steatosis assessment in liver transplantation further reinforces this concept (Table 1).

Regardless, we agree that a larger sample size is always preferable. To this end, we are collaborating with international institutions to extend and further validate the results of our study.

Our cohort of liver biopsies was also questioned due to a high prevalence of cases with low steatosis (per authors’ letter: “the majority (75%) contain fat content below 30%.”). Unfortunately, our results may have been misunderstood. Our study reported the steatosis percentage as mean ± standard deviation. We used semi-quantitative cut-offs (minimal: <1%, mild: 1-10%, moderate: 10-30%, and severe: >30%) to report the difference between the percentages provided by the pathologists and the convolutional neural network models. Indeed, this difference in score (but not the steatosis percentage *per se*) was below 30% in most cases. We apologize for any misunderstanding this approach caused, but we would also like to discuss the reason behind this choice. To date, most literature data reported steatosis percentage using semi-quantitative cut-offs as the ones

suggested by the authors (mild: <30%, moderate: 30-60%, severe: >60%).^{11,12} However, we observed that Banff recommendations generally provided lower percentages of steatosis. Kikuchi and colleagues recently confirmed this trend: they compared the Banff recommendations with another scoring system (*i.e.*, the “Average of Fields” method) in 10 donor liver biopsies and observed that steatosis percentages were significantly lower when adopting Banff criteria.¹³ As a consequence, cases that were reported as moderate (30-60% of steatosis) or severe (>60% of steatosis) before Banff criteria implementation may now risk being inappropriately classified as mild (<30%) if we maintain the same thresholds. Being aware of this caveat, we preferred to avoid potentially misleading classification pending the definition of more representative cut-offs.

Another concern was the analysis of clinical data and correlations with transplant outcomes. Indeed, our study focused on how the assessment of donor steatosis changed with Banff recommendations and artificial intelligence approaches, as we additionally discussed above. We agree that a thorough correlation with clinical data would be extremely meaningful, and, as we pointed out in the discussion of our study, we are currently exploring the effect of Banff recommendations on the clinical outcomes of our cohorts of patients.

Finally, it was suggested that more information on our deep-learning model should be provided and its performances compared with traditional computer vision techniques and other deep-learning approaches. Regarding the availability of information on our model, the codes of our GAN for stain normalization are open-access and publicly available (doi: 10.17632/32bvf6xhj.2), as well as the intermediate results of our study (steatosis segmentation, tissue segmentation, pre-Banff approach, and Banff method; <https://data.mendeley.com/datasets/cjgd4wr2tz/1>), and the dataset and manual annotations of our previous work in this field (<https://data.mendeley.com/datasets/4mcc9rg4k5/1>).¹⁴ We fully support and encourage open science, and we invite investigators to access our content and contact us with any further requests or questions. The proposition to test and compare different digital pathology tools is fascinating, and we thank the authors for the suggestion. We agree that traditional computer vision

Table 1. Recent studies on liver donors’ steatosis assessment via artificial intelligence and related sample size.

Study	Publication year	Sample size (number of liver biopsies analyzed)	Reference
Lee <i>et al.</i>	2013	Overall biopsies: 41; Liver transplant-only biopsies: 18	5
Sun <i>et al.</i>	2020	96	6
Cesaretti <i>et al.</i>	2020	117	7
Narayan <i>et al.</i>	2021	90	8
Balanza <i>et al.</i>	2023	Overall sample size (patients): 350; With available liver biopsy: 123	9
Tang <i>et al.</i>	2024	109	10
Our study	2024	292	2

techniques and related simple thresholding approaches are still relevant in specific settings. However, they are also prone to identify and consider irrelevant portions of the tissue, such as partial fragmentation, dilated sinusoids, and other non-target structures.¹⁴ Deep learning-based methods proved to be superior in histopathology image analysis and liver steatosis segmentation,^{14,15} and we agree that it would be interesting to compare our deep-learning method with other similar approaches. Hence, we will soon start exploring this topic further.

In summary, future studies addressing artificial intelligence in donor steatosis assessment should focus on databases representative of the liver transplant population. We envision an important effect of Banff steatosis recommendations and deep learning techniques on liver transplantation and donor organ eligibility, and we may need to embrace new, more representative cut-offs as we move the field forward.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors' contributions

AG and MS provided acquisition, analysis, and interpretation of data, and wrote the original draft; FM supervised manuscript development. All authors read and approved the final version of the manuscript.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.05.015>.

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Author names in bold designate shared co-first authorship

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